SET-A University Roll No._____

Term I Examination, 2018-19
B. Tech. II Year (Even Semester)

Subject: English for Professional Purposes–II **Subject Code: BELH 0004**
Time: 60 Minutes **Max. Marks: 15**

Note: i. It's mandatory to Cleary mention your set no in answer sheet.
ii. Write only the options (such as A, B, C etc.) in the answer sheet while answering Verbal Aptitude questions.

PART A - ORAL SKILLS **TIME: 30 MINUTES**

Section- A **0.5x4 = 2 Marks**

Note: Attempt all the questions.

1. Multi-syllabic words ending with –cy, -ity, -gy, -phy, etc. receive stress on _____ syllable.
2. What information does the first paragraph of a cover letter ideally contain?
3. Read the sentence given below and identify the tone of the speaker.
   'Don't go out.' (Command)
4. It is advisable to spend some time to look at the demographics of the audience while preparing a presentation. True/False

Section- B **1.5x2 = 3 Marks**

Note: Attempt any two questions. (In not more than 60 words)

5. Write any 3 rules for putting stress on a particular syllable of a word.
6. You are a graduate applying for a job. Write your educational qualification in the resume.
7. Write any three characteristics of an effective group presentation.

Section- C **2.5x1 = 2.5 Marks**

Note: Attempt any one question. (In not more than 125 words)

8. You have earned your M Tech degree and are seeking for a teaching position at a university. Draft a cover letter.
9. Break the following words into syllables and mark stress on appropriate syllable.

Technology,	Ability,	Political,	Engineer,	Student,
Relation,	Excellency,	Autocracy,	University,	Ago

PART B- VERBAL APTITUDE **TIME: 30 MINUTES**

Note: Attempt all the questions. Each question carries 0.25 marks.

Mark the most appropriate choice as your answer. (Q 1-12)

1. You can't blame ____ me for the feeling that it is an end of the era.
   A. at B. for C. on D. to E. No prep. required
2. It is very cold and foggy weather. You may fall ill if you go outside. The underlined word in the sentence is a _____.

A. preposition B. adverb
C. adjective D. conjunction
E. none of these

3. "The managing director of the bank is in the firing line since the fraud was discovered." The underlined part in the sentence means _____.
   A. he is a fireman
   B. he is a soldier
   C. he is in a position to be criticized
   D. he is being scolded
   E. none of these
4. "Mr Sinha has been forced to retire or give up his responsibilities." Which of the idioms given below can express a meaning suitable for the underlined part in the sentence?
   A. put out to pasture
   B. pound the pavement
   C. plum job
   D. paid peanuts
   E. duty bound
5. "The rumour of Professor Agrawal's appointment as Chief Secretary of the State became widely known very fast." In the sentence given above, which of the following idioms can suitably be used?
   A. put someone in picture
   B. speed networking
   C. spread like wildfire
   D. hit the airwaves
   E. keep someone posted
6. When you hear through the grapevine, you _____.
   A. are intoxicated
   B. are using hearing machine
   C. learn about it informally
   D. are informed about it through proper channel
   E. none of these
7. You will be awarded with an excellence certificate if you _____ extraordinarily.
   A. would have performed
   B. are performing C. perform
   D. will perform E. performed
8. Would you mind if I _____ you to stay with us for some more days?
   A. would request B. request
   C. was request D. requested
   E. would have requested
9. They would have left if they _____.
   A. should have had to B. had to
   C. had had to D. could
   E. could have
10. She would be nothing but a fool, if she _____ his advice.
    A. followed B. would follow
    C. will follow D. follows
    E. had followed
11. Rob _____ English if he hadn't gone to class.
    A. would learnt B. will not learn
    C. would not have learnt
    D. would not have learn
    E. would not learn
12. Dr Mehrotra is fond ____ music.
    A. of B. at C. from
    D. about E. with

Identify the erroneous part of the sentence. (Q 13-16)

13. The affiliation process of several new schools (A)/ too the Uttar Pradesh Board has been delayed (B)/ due to non submission of report by (C)/ the District inspector of schools of different states (D)/ No Error (E)
14. The district jails in the state are overcrowded by (A)/ prisoners and no effective measures (B)/ are being taken (C)/ to improve the infrastructure (D)/ No Error (E)
15. He climbed the ladders of success (A)/ by the dint of hard work (B)/ and earned a reputation of (C) being a self made person (D)/ No Error (E)
16. Caste, cash and community must not be allowed (A)/ to play any role in electoral politics (B)/and people must elect their public representatives(C)/

on the basis of character, calibre, capacity and conduct (D)/ No Error (E)

Do as directed (Q 17-20):

17. Identify the correct expression:
    A. If winter came, can spring be far behind!
    B. If winter comes, could spring be far behind!
    C. If winter comes, can spring have been far behind!
    D. If winter comes, can spring be far behind!
    E. If winter will come, can spring be far behind!
18. Identify the correct expression:
    A. He would have been here earlier if he had not missed the bus at 6.
    B. He would have here earlier if he had not missed the bus at 6.
    C. He would be here earlier if he had not missed the bus at 6.
    D. He would have been here earlier if he was not missed the bus at 6.
    E. None of the above is correct.
19. Which of the pair of conditional clause and the verb-form is correct?
    A. Zero conditional –Simple present
    B. First conditional- Would+V_1
    C. Second conditional – Simple past
    D. Third conditional- past perfect
    E. None of these
20. Which of the following statements about conditional sentences is not true?
    A. A conditional sentence is a complex sentence.
    B. A conditional sentence has at least two dependent clauses.
    C. Among the types of conditional sentences, first and second conditionals talk about future possibilities- one real and the other, unreal.
    D. Zero conditional describes scientific facts.
    E. A conditional clause does not necessarily begin with 'if'.

Write the meanings of the following idioms and frame sentences using them in a suitable context. (Q 21-26)

21. Drop someone a line
22. Word of mouth
23. Worth one's salt
24. Chinese whisper
25. Back-room boys
26. Golden handcuffs

Read the passage and answer the questions that follow. (Q 27-30)

It is true that some people contend that economic growth is the only solution to the global problems of hunger and poverty. Others, however, argue for an end to economic growth in order to conserve our environment. I agree completely with this second view.

Those who see economic growth alone as the sole cure for the tragedy of world hunger and poverty propose one major argument. Only the growth of the economies of the developing countries will provide the poor with the wealth to afford the basic necessities of life. The profits made by corporations who are responsible for this economic growth will trickle down in the form of financial benefits to be enjoyed by the starving and needy. This view has justified the age of imperialism and the destruction of the livelihood of millions in the name of progress.

On the other hand, there are countless examples to support the opposing view that economic growth results in environmental destruction. Firstly, this is true for developing countries, where lowland rainforests have been cleared and unsustainable agricultural practices

introduced to produce cash crops, often for export. Secondly, it is also true for developed economies. For example, the continued use of fossil fuels, which provide the energy for growing industrialisation, is causing global warming. This is leading to a rise in sea levels, which will eventually make tens of millions of poor people in Bangladesh homeless.

In conclusion, economic growth which is not sustainable must be stopped. If it is not based on meeting human needs rather than generating profits, such growth will only create more poverty and lead to more hunger in the world by destroying natural resources.

27. Which of the following words used in the passage mean the same as "to say that something is true especially in an argument"
    A. Corporate
    B. Contend
    C. Cause
    D. Sustain
    E. None of these
28. According to those who find economic growth alone as the cure for hunger and poverty,
    A. only the economic growth of the of the developing countries can help the poor afford the basic necessities of life.
    B. the profits made by corporations won't trickle down in the form of financial benefits to be enjoyed by the starving and needy.
    C. the view never justifies the age of imperialism and the destruction of the livelihood of millions in the name of progress.
    D. economic growth results in environmental destruction.
    E. None of the above is correct.
29. Which of the statements given below does not support the view: "Economic growth results in environmental destruction"?
    A. In developing countries, lowland rainforests have been cleared and unsustainable agricultural practices introduced to produce cash crops, often for export.
    B. For developed economies, the continued use of fossil fuels provides the energy for growing industrialisation, which is causing global warming.
    C. Growing industrialisation is leading to a rise in sea levels, which will eventually make millions of poor people homeless.
    D. All support the statement.
    E. None supports the statement.
30. The passage concludes that:
    A. economic growth which is not sustainable must not be stopped.
    B. economic growth which is not sustainable must be stopped.
    C. If it is based on meeting human needs rather than generating profits, such growth will only create more poverty.
    D. Both A and C are correct.
    E. Only B is correct.

SET-B University Roll No._____

Term I Examination, 2018-19
B. Tech. II Year (Even Semester)

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PART A - ORAL SKILLS TIME: 30 MINUTES

Section- A 0.5x4 = 2 Marks

Note: Attempt all the questions.

1. Multi-syllabic words ending with –cy, -ity, -gy, -phy, etc. receive stress on _____ syllable.
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4. It is advisable to spend some time to look at the demographics of the audience while preparing a presentation. True/False

Section- B 1.5x2 = 3 Marks

Note: Attempt any two questions. (In not more than 60 words)

5. Write any 3 rules for putting stress on a particular syllable of a word.
6. You are a graduate applying for a job. Write your educational qualification in the resume.
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Section- C 2.5x1 = 2.5 Marks

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PART B- VERBAL APTITUDE TIME: 30 MINUTES

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Mark the most appropriate choice as your answer. (Q 1-12)

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    D. Both A and C are correct.
    E. Only B is correct.

Write the meanings of the following idioms and frame sentences using them in a suitable context. (Q 25-30)

25. Drop someone a line
26. Word of mouth
27. Worth one's salt
28. Chinese whisper
29. Back-room boys
30. Golden handcuffs

Paper Code-BTDH 0302

BTDH0302

SET-A

Soft Skills

First Mid Term, Even Sem Examination (2018-19)

Course- B.tech/MCA Year- II, I

Branch- (All)

Time : 60 Min. M.M-20

Instructions :

1. Attempt all questions.
2. There will be no negative marking. Also, no marks will be awarded if more than one answer is marked.
3. The OMR form is printed on a special paper and the information given by the candidates will be scanned. Thus, the candidates are required to fill the forms carefully. Mutilated and damaged forms will be rejected by the scanner.
4. The candidates should ensure that the form is not folded, crumpled or mutilated in any manner. The candidates should not mark anything anywhere except wherever required. The black marks on the margin of the OMR form are important. The candidates should not tamper it with marker.
5. Only black& blue pen should be used to fill forms.
6. Each branch is identified with a specific subject code which should be written in subject column of OMR sheet
7. Write your University roll number in the roll no column of OMR sheet.

Name of Student:-.....

Branch:-.....

Section-.....

Uni. Roll No:..... All The Best

(Aptitude)-No of question-40, marks-0.50

Q1. What is the angle covered by the minute hand in the same time in which hour hand cover 14^0?
A. 168° B. 14° C. 220° D. None of these

Q2. Calendar of 1900 will be repeated in for the first time?
A. 2006 B. 1906 C. 1928 D. None of these.

Q3. 20% of 20 is what percent less than by 40% of 40:
A.25% B. 50% C.75% D. None of these

Q4. Selling price of an article is 20% more than its cost price. After giving a discount of 10%, what is the profit gained by shopkeeper?
A. 10 % B.20 % C. 8 % D. None of these

Q5. Compound interest for the 2nd year on a sum of money is Rs 132 at the rate of 10% per annum compounded annually. The sum is
A. 1000 B. 1200 C. 1500 D. None of these

Q6. The rate at which a sum becomes 5 times of itself in 25 years at S.I., will be?
A.8 % B.4 % C.16% D. None of these

Q7. If 2123XY is divisible by 33. How many solutions for (X, Y) are possible?
A.0 B.1 C.3 D. None of these

Q8. What is the mirror image time of 15 minutes past 9 O'clock?
A. 3:45 B. 2:45 C. 1:45 D. None of these

Q9. Which of the following is not a leap year?
A. 2012 B.1600 C. 1900 D. 7200

Q10. The price of an article first increases by 20%, again increases by 30% and then decreases by 50% in the third year. What will be its final value, if its initial price is Rs. 10,000?

A. 7200 B. 7600 C. 7800 D. None of these

Directions (11-13): Study the following information carefully to answer the questions that follow: A is father of B and C is mother of A. E is sister of F whose daughter is G. S, the husband of C is the grandfather of G. P is father of E and brother of R. S has only two children, both of opposite sex.

Q11. What is the relation between F and S?

A. F is daughter of S B. F is sister of S C. F is son of S

D. F is daughter in law of S E. Either A option or D

Q12. What is the relation between E and B?

A. E is sister of B B. E is brother of B C. E is aunt of B

D. E is maternal grandmother of B E. None of these

Q13. What is the relation between B and G?

A. B is sister of G B. B is brother of G C. B is aunt of G

D. There is no relation E. None of these

Directions (14 – 15): Consider the following relationships and answer based on the relationships given below:

'P % Q' means 'P is daughter of Q'

'P # Q' means 'P is wife of Q'

'P $ Q' means 'P is father of Q'

'P @ Q' means 'P is brother of Q'

Q14. N @ B $ M % D indicates what relationship between N and D?

A. D is sister of N B. D is nephew of N C. D is sister in law of N

D. N is wife of D

E. None of these

Q15. The expression "M # Q @ D % S" indicates which of the following relationship?

A. Q is father of S B. D is brother-in-law of M C. M is sister of S

D. S is mother-in-law of M E. None of above

Q16. Micael said, pointing to Mahavir "He is son of my grandfather's daughter". How is Micael related to Mahavir?
A. Grandfather B. Cousin C. Uncle D. Brother
E. None of these

Q17. Pointing to a lady in photograph, Nitesh said, "Her mother is the only daughter of my mother's mother." How is Nitesh related to the lady?
A. Nephew B. Uncle C. Maternal uncle D. Brother
E. None of these

Q18. P, Q, R, S, T, U, V, W are sitting around a round table in the same order, at equal distances. Their positions are anticlockwise. If V sits in the North. Then what will be the positions of S?
A. East B. South East C. South D. South West
E. cannot be determined

Q19. One morning after Sunrise. Riya & Lovely were talking to each other face to face at Tilak Square. If the Lovely's shadow was exactly to the right of Riya, which direction Lovely was facing?
A. North B. South C. East D. Data Inadequate
E. None of these

Q20. B is to the South- West of A, C is to the East of B and South East of A, D is to the North of C in the line with B and A. In which direction of A is D located?
A. North B. East C. South- East D. North- East
E. None of these

Q21. A watch Read 4:30, If the minute hand points east. In which direction will the Hour hand point at 10:30?
A. South- East B. South West C. North D. North West
E. None of these

Q22. The town of Paranda is located on Green Lake. The town of Akram is west of Paranda. ToKhada is East of Akram but west of Paranda. Kokrein is East of Bopri but west of Tokhada and Akram. If they are all in the same district which town is the farthest west?
A. PARANDA B. KOKRAN C. AKRAM D. BOPRI
E. None of these

Q23. Ms. A goes for her morning walk at 6 'O' clock towards Sun for 2km Then she turns to her right and walks 3km. She again turns to her left and walks 2km. Finally, she turns to her left to walk another 6km. In which direction is she moving and at what distance from starting point?
A. 6km North-East B. 5km North-East C. 6km North-West
D. 9km North-West E. None of these

Q24. A person starts towards South direction. Which of the following orders of directions will lead him to East direction?
A. Right, Right, Right B. Left, Left, Left C. Left, Right, Right
D. Right, Left, Right E. None of these

Q25. If a: b = 3: 4 and b: c = 7: 9, then find a: b: c.
A. 3:28:9 B. 3:28:36 C. 21:28:36 D. None of these

Q26. If Suresh distributes his pens in the ratio of 1/2: 1/4: 1/5: 1/7 between his four friends A, B, C and D, then find the total number of pens Suresh can have?
A. 153 B. 306 C. 459 D. All of the above

Q27. The annual income of Puja and Jaya taken together is Rs. 44,000. Puja spends 90 % of income, and Jaya spends 80% of her income. If their annual savings are in the ratio of 3: 2, find the annual income of Puja?
A. 11, 000 /- B. 12, 000 /- C. 17, 500 / - D. None of these

Q28. A bag contains equal number of 25 paise, 50 paise and one rupee coins respectively. If the total value is Rs 105, how many coins of each type are present?
A. 75 coins B. 60 coins C. 30 coins D. 25 coins

Q29. A wheel that has 6 cogs is meshed with a larger wheel of 12 cogs. If the smaller wheel has made 22 revolutions, then find the number of revolutions made by the larger wheel.

A. 11 B. 13 C. 15 D. 17

Q30. 18 men bind 900 books in 10 days. Find how many binders will be required to bind 600 books in 12 days?

A. 10 B. 11 C. 13 D. 15

Q31. If $x^2+4y^2=4xy$, then x: y is:

A. 1:2 B. 2:1 C. 2:3 D. 3:2

Q32. The sub-triplicate ratio of 27:64 is:

A. 1:2 B. 2:1 C. 3:4 D. 4:1

Q33. John, Tyson and Mike started a business by investing Rs.1,00,000, Rs. 1,50,000 and Rs. 1,75,000 respectively. Find the share of Mike, out of an annual profit of Rs. 46,000.

A. Rs. 15941. 93 B. Rs. 16943. 59 C. Rs. 18941. 17
D. Rs. 19971. 03

Q34. Rohan began a business with Rs. 85,000 and was joined by Sanket with Rs. 42,500. If the profits at the end of the year are divided in the ratio of 3: 1, then find for how much period of time did Sanket join Rohan?

A. 10 months B. 8 months C. 7 months D. 6 months

Q35. Prakash, Sunil and Anil started a business jointly investing Rs.11 lakhs, Rs. 16.5 lakh and Rs. 8.25 lakhs respectively. The profit earned by them in the business at the end of three years was Rs.19.5 lakhs. What will be the 50% of Anil's share in the profit?

A. Rs. 4.5 lacs B. Rs. 2.25 lacs C. Rs. 2.5 lacs
D. None of these

Q36. Find the average of all numbers between 5 and 49 which are divisible by 5.

A. 20 B. 22 C. 25 D. 35

Q37. The averages of 11 numbers is 30. If the average of first six numbers is 17.5 and that of last six is 42.5, then what is the sixth number?

A. 30 B. 36 C. 45 D. 47

Q38. The average of four consecutive even numbers is 27. Find the largest of these numbers.

A. 28 B. 30 C. 32 D. 34

Q39. There are two batches A and B of a class. Batch A consists of 50 students and batch B consists of 70 students. Find the average weight of whole class, if average weight of batch A is 40 kg and that of batch B is 35 kg.

A. 39.23 kg B. 32.56 kg C. 35.66 kg D. 37.08 kg

Q40. The average of 20 numbers is zero. Of them, at the most, how many may be greater than zero?

A. 0 B. 1 C. 10 D. 19

First Mid Term Examination Even Semester 2018-19

Programme: B.Tech Branch: CSE Year: II

Subject with Code: Data Structures & Algorithms, BCSC 0006

Time : 1 Hour **Maximum Marks: 15**

Section – A

3 x 2 = 6 Marks

Q1. What are Data Structures? Explain operations that can be performed on data structures.

Q2. Explain the criteria that you will keep in mind while choosing an appropriate algorithm to solve a particular problem.

Q3. What is ADT. Give Example.

Section – B

3 x 3 = 9 Marks

Q1. Write a program to create a queue using arrays which permits insertion and deletion of Book Class object that is having two attributes (Name, Price).

Q2. What do you understand by stack and queue? Explain in detail

Q3. Discuss the significance and limitations of the Big 0 notation.

I Mid-Term Examination

Even Semester, 2018-19

Programme: B. Tech. Branch: CSE/CCV/DA Year: II

Subject with Code: Discrete Mathematics, BCSC0010

Time : 1 Hour **Maximum Marks: 15**

Section – A 3 x 2 = 6 Marks

1. A survey of 500 television watchers produced the following information: 285 watch football games, 195 watch hockey games, 115 watch basketball games, 45 watch football and basketball games, 70 watch football and hockey games, 50 watch hockey and basketball games and 50 do not watch any of these games.
a) How many people in the survey watch all three kinds of games?
b) How many people watch exactly one of the sports?

2. Prove the following:
A X (B – C) = (A X B) – (A X C)

3. a) Find the closed form of generating function for the following sequence:
1, -2, 3, -4, ...

b) What do you understand by a bijective function?

Section – B 3 x 3 = 9 Marks

1. If R is a relation in the set of natural numbers N defined by "x-y is divisible by 7" i.e. R = {(x,y) : x,y ∈ N, (x-y) is divisible by 7}, prove that R is an equivalence relation.

2. Solve the following recurrence relation:

$$a_n - 5a_{n-1} + 6a_{n-2} = 2.3^n \text{ where } a_0 = 1 \text{ and } a_1 = 3$$

3. Attempt the following: (1+2)

a) If there are several black, white and green balls in a bag, then find minimum number of balls that must be chosen to be sure of getting two pairs of balls with the same color.

b) In how many different ways can the letters of the word 'SALOON' be arranged for the following cases?
i) Two O's must not come together.
ii) Consonants and vowels must occupy alternate places.

I Mid-Term Examination Even Semester, 2018-19

Programme: B.Tech. **Branch: CS** **Year:2nd year**

Subject with Code: Computer Networks, (BCSC0008)

Time : 1 Hour **Maximum Marks: 15**

Section – A

Attempt all questions. 3 x 2 = 6 Marks

1. a) 50 employees are working in an organization and everybody want to link with everyone. If the length of one link is 5 meters, then how much length of cable required?
   b) What is difference between bit rate and baud rate? Explain it with suitable diagram?
2. a) What is vulnerable time in slotted aloha?
   b) A sine wave is offset 1/6 cycle with respect to time 0. What is its phase in degrees and radians?
3. a) Suppose two friends have created a network using Bluetooth. Mention the name of IEEE standard that these friends will be making use of.
   b) What is the relationship between period and frequency?

Section – B

Attempt all questions. 3 x 3 = 9 Mark

1. How pure ALOHA differs from slotted ALOHA? One hundred stations on a pure ALOHA network share a 1-Mbps channel. If frames are 1000 bits long, find the throughput if each station is sending 10 frames per second.
2. a) Compare and contrast a circuit-switched network and a packet-switched network.
   b) How do guided media differ from unguided media?
3. A 2 km long broadcast LAN has 10^7 bps bandwidth and uses CSMA/CD. The signal travels along the wire at 2×10^8 m/s. What is the minimum packet size that can be used on this network?

I Mid-Term Examination
Even Semester, 2018-19

Programme: B.Tech Branch: CSE Year: II
Subject & Code: Introduction to Microprocessor (BCSC0007)
Time: 1 Hour Maximum Marks: 15

Section – A 3 x 2 = 6 Marks

Q1. Define Microprocessor? What do you understand with word size of microprocessor? Distinguish between microprocessor and microcontroller (Any two)?

Q2. Enlist functional block of 8085 and describe Instruction decoder & Machine cycle encoder unit in brief.

Q3. The instruction MOV A, B is useful for copies the contents of register B into register A. It is a 1-Byte instruction. Draw the timing diagram for the machine cycle for MOV A, B.

Section – B 3 x 3 = 9 Marks

Q1. Enlist, how can signals be classified for the 8085 Microprocessor? Explain the use of Ready, HOLD and A_{15}-A_8 signals in 8085.

Q2. What is Instruction Format? The 8085 can perform many instructions, these instructions can be functionally classified. Enlist the classification and explain any one of them.

Q3. Write instruction used for given descriptions with its size (in bytes) and addressing mode.

a. Load 2200H in any register pair.
b. Subtract the contents of Register B with the contents of accumulator with borrow
c. Clear the contents of accumulator
d. Load the contents of memory location 2000H in accumulator.

First Mid Term Examination, 2018-19
Course: B.Tech. (CSE), Year:II, Semester:IV
Software Engineering (BCSC 0009)

Uni. Roll No: _____

Time: 1 Hrs. **Total Marks: 15**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(3x2=6)**

I. "Is Requirement gathering a most error prone and most communication intensive process". Justify using proper comments. In your opinion which is the best method to take all types of requirements from client and why?

II. Explain the need of Use case diagram? Suppose we want to develop software for an alarm clock. The clock shows the time of day. Using buttons, the user can set the hours and minutes fields individually, and choose between 12 and 24-hour display. It is possible to set one or two alarms. When an alarm fires, it will sound some noise. The user can turn it off, or choose to 'snooze'. If the user does not respond at all, the alarm will turn off itself after 2 minutes. 'Snoozing' means to turn off the sound, but the alarm will fire again after some minutes of delay. This 'snoozing time' is pre-adjustable. Identify the top-level functional requirement for the clock, and model it with a use case diagram.

III. A software company needs to develop a project that is estimated as 1000 function points and is planning to use JAVA as the programming language whose approximate lines of code per function point is accepted as 50. Considering a = 1.4 as multiplicative factor, b = 1.0 as exponential factor for the basic COCOMO effort equation and c = 3.0as multiplicative factor, d = 0.33

as exponential factor for the basic COCOMO duration equation, Apply COCOMO Model and Identify approximately how long does the project take to complete, average staff size required to complete this software on time.

Section B

Note: Attempt All Questions **(3x3=9)**

I. Compare the Levels of DFD. Create Level-0, Level-1 and Level-2 DFD for online shopping website. The admin of the website can login as an admin and can manage the items of the company. Admin can manage the orders placed by customers, generates the monthly sales report. Users can login, search, place order, track order and online payment of purchased items.

II. A project is assigned to a software industry by a client. Which model should be used for software development under which requirements and solutions evolve through the collaborative effort of self-organizing and cross functional teams where team members size must be 3 -8 members with project leader and team members must have a meeting for 15 minutes on day to day basis. Define the process of this model by explaining all the components with suitable diagram.

III. Identify the effort and Duration of a project with the following domain characteristics:

No. of I/P = 30

No. of O/P = 62

No. of user Inquiries = 24

No. of files = 8

No. of external interfaces = 2

Assume that all the complexity adjustment values are average. Assume that 14 algorithms have been counted. The software is going to be implemented in C++ language.

Printed Pages:1 **University Roll No.....**

MCA- I Year (II-Semester)
First Term-I Exam (Even)
Title of Paper: Environmental Studies
Paper Code: (BCHS-0201)

Time: 01 Hours. **Max. Marks: 15**

Section – A

Note: Attempt All Questions. **[2x3=6 Marks]**

Q1. Discuss the contribution of environmental science in discovering the solution to environmental problems.

Q2. State the effect of different abiotic factors that influences the sustainability of any ecosystem.

Q3. Suggest some effective measures for the conservation of natural resources.

Section – B

Note: Attempt All Questions. **[3x3=9 Marks]**

Q1. Discuss the vertical structure of atmosphere with the help of a labeled diagram, stating all important activities taking place in different temperature zone regions.

Q2. Explain the following term

A.) Intangible Natural Resources. B.) Coliform number count.

C.) Ecological Succession.

Q3. Enlist the possible ways by which mineral resources can contribute to the economic development of any country.